

NACA Airfoil & Boundary Layer Simulator

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Application URL: <https://poompm.github.io/Airfoil-simulator-vibe/>

1 1. Overview and Description

The Advanced NACA Aerodynamics & Boundary Layer Simulator is a fully interactive, web-based engineering tool designed to model external incompressible viscous flow over standard NACA 4-digit airfoils. Built using HTML5 Canvas and advanced vector field mathematics, the application bridges theoretical fluid mechanics with real-time visual feedback. It simulates complex aerodynamic phenomena including laminar-to-turbulent boundary layer behavior, adverse pressure gradients, surface flow tangency, and aerodynamic stall.

2 2. Core Features & Technical Capabilities

- **Dynamic NACA 4-Digit Generation:** Users can customize airfoil profiles in real time by adjusting the Maximum Camber (M), Camber Position (P), and Thickness (T), instantly rendering exact geometry using standard aeronautical equations.
- **Rigorous Fluid Physics Engine:**
 - **No-Slip Boundary Condition:** Simulates fluid velocity decay near the solid surface.
 - **Surface Tangency:** Streamlines mathematically align with local camber slopes to naturally hug the airfoil contour.
 - **Separation and Wake Modeling:** Automatically computes the separation point ($\partial u / \partial y|_{y=0} = 0$) and models chaotic eddies within the low-pressure wake zone during a stall.
- **Dual Visualization Modes:** Switch seamlessly between Streamline Particles (motion-blurred moving tracer particles) and Continuous Lines (steady-state vector pathlines).
- **Live Telemetry & Metrics:** Real-time computation of the Lift Coefficient (C_L), Drag Coefficient (C_D), and flight status indicators (Attached Flow vs. Stall).
- **Dynamic Infographic Annotations:** On-screen visual indicators automatically track and point to critical aerodynamic regions, including the Stagnation Point, Viscous Boundary Layer, Separation Point, and Turbulent Wake.

3 3. User Guide & Instructions

3.1 Step-by-Step Operation

1. **Access the Application:** Open the published web link in any modern web browser: <https://poompm.github.io/Airfoil-simulator-vibe/>
2. **Select Visualization Mode:** Use the dropdown menu at the top of the control panel to choose between Streamline Particles (best for viewing dynamic flow momentum) or Continuous Lines (best for steady-state field patterns).
3. **Configure Airfoil Geometry:**
 - **Max Camber (M, %):** Adjusts the first digit (maximum camber as a percentage of chord).

- **Camber Position (P, %):** Adjusts the second digit (location of max camber along the chord).
- **Thickness (T, %):** Adjusts the final two digits (maximum thickness profile as a percentage of chord).

4. Adjust Flight Conditions:

- **Angle of Attack (α):** Use the slider to pitch the airfoil nose up or down relative to the relative wind (α is measured positive counterclockwise/clockwise relative to the horizontal wind vector). Push past $\sim 13.5^\circ$ to trigger an aerodynamic stall.
- **Relative Wind Speed:** Adjusts the free-stream velocity magnitude (U_∞) of the flow field.

5. Analyze Telemetry & Infographics:

- Observe how the red separation point indicator moves forward along the upper surface as the Angle of Attack increases.
- Monitor live changes in C_L and C_D within the metrics panel.